PORTFOLIO 2026

Chris Al-Rahi

Mechanical Engineering
& Computing Technology
at uOttawa

Co-op · Graduating April 2027 · Ottawa, ON



ABOUT ME

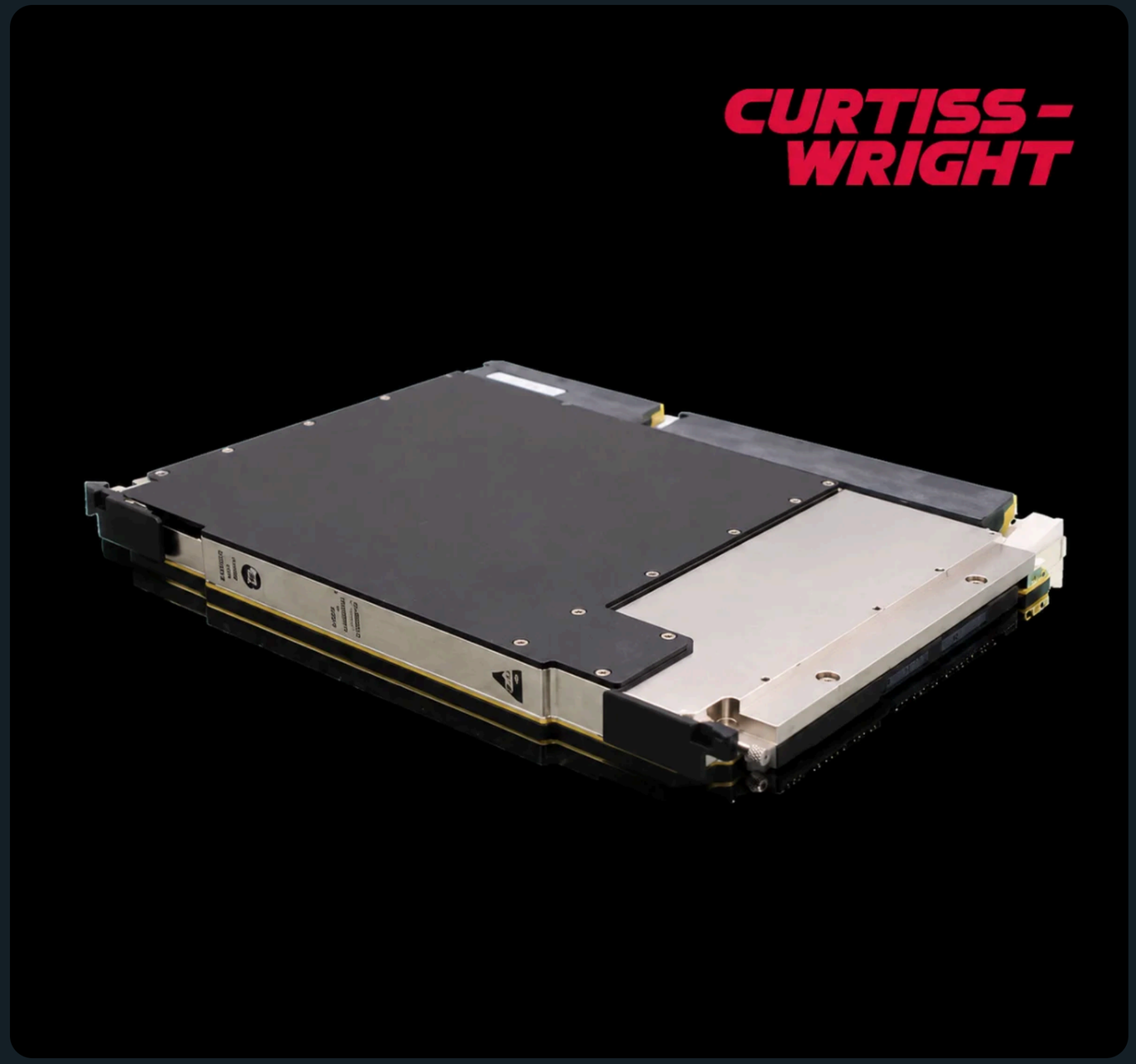
Building is my passion. Understanding is my method.

School design projects, household repairs, prototypes I want to test in SolidWorks - I go looking for problems worth solving. When something fails, I take it apart, find out why, and rebuild it better.

I'm competitive, and I commit fully to what I take on. Right now that means a six-axis robot arm on the bench, piano in the evenings, and training for my next fight.

Muay Thai | Scuba Diving | Volleyball | Travel | Reading





WORK · CURTISS-WRIGHT · 2025

Liquid-Cooled Chassis Redesign

I reworked the liquid cooling geometry of a ruggedized electronics part to see if cheaper machining methods could hold the same thermal performance.

MY ROLE Redrew the cooling paths in CAD around cheaper machining processes, then ran Flotherm simulations to compare the new geometry against the original.

OUTCOME A manufacturable, lower-cost layout the team could build against, alongside thermal and environmental testing on nine conduction-cooled products.

Flotherm · Thermal design · CAD

WORK · CANMETENERGY · RH&P · 2025

Aspirated & Shielded RTD Sensor

Air temperature readings inside the SolarSteam enclosure were being thrown off by radiation, so I built a shielded sensor that slides up and down to sample at any height.

MY ROLE Designed the shield in SolidWorks, sized the aspiration from the literature, and ran the accuracy testing myself.

OUTCOME Live temperature accuracy improved by more than 30%, and I presented the work as a research poster.

SolidWorks · Instrumentation · Field testing



CO-FOUNDER INFYLWORKS.COM

Infyl

A custom parts business my friend Nik and I started. People tell us what they need, and we design the part in CAD from their measurements, 3D print it, and ship it.



Dragon Echo Holder

A sculpted stand that wraps a smart speaker.



Custom Storage System

Organizers sized to a customer's furniture. One example shown.



Dual Google Home Mini Holder

Two speakers mounted on the side of a vanity.

PERSONAL · WITH NIK · IN PROGRESS

Six-Axis Robot Arm

A desktop arm my friend Nik and I are building to lift finished prints off the 3D printer on its own. Six stepper motors drive planetary reducers we print ourselves, with a camera on the wrist.

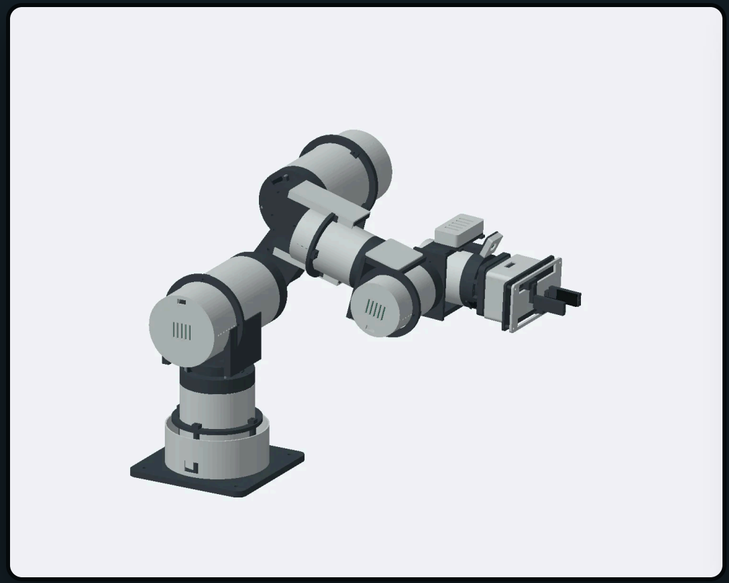
MY ROLE

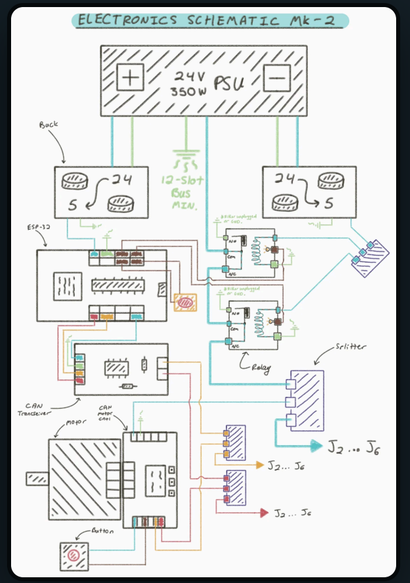
Nik and I worked together to make different parts of the mechanical design, iterating on each other's revisions of the electronics schematic and the building plan.

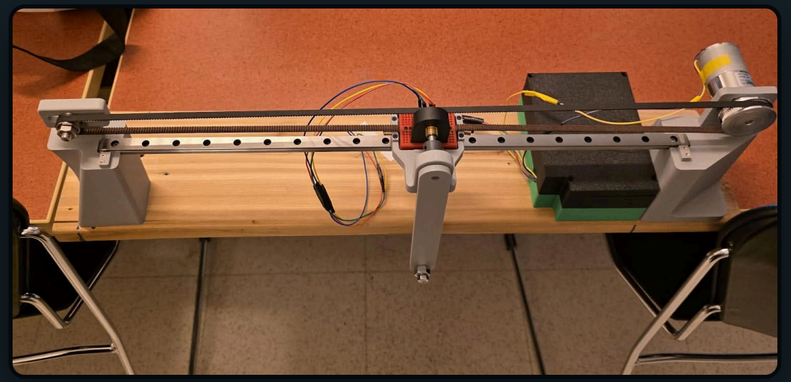
STATUS

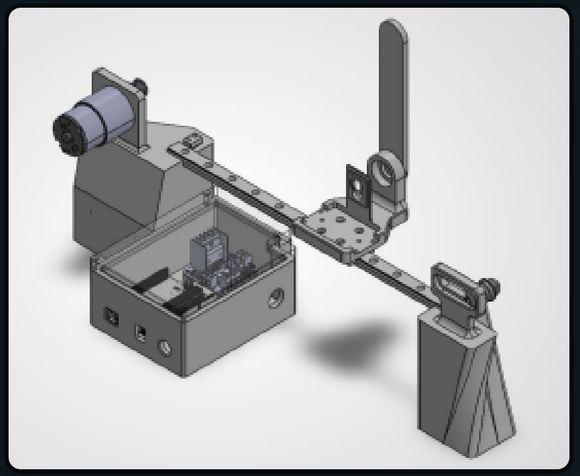
A complete CAD prototype with 93 printed parts, the wiring schematic, and every motor, bearing and board chosen. Next: print the first reducers and test them on the bench.

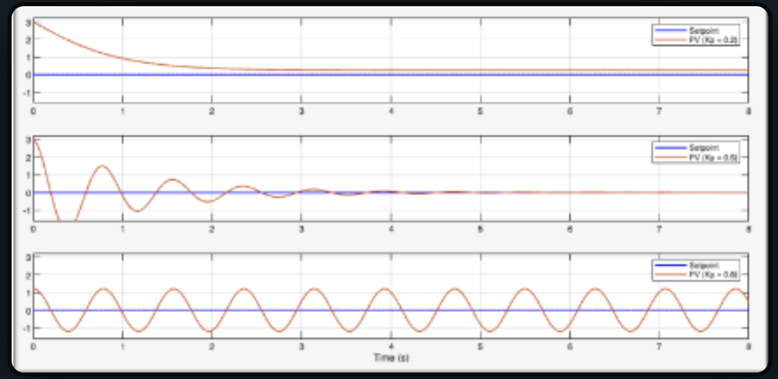
3D printing · Planetary gearing · ESP32 + CAN bus · Stepper motors











SCHOOL CONTROL SYSTEMS

Inverted Pendulum

A belt-driven cart on a 0.5 m rail that balances a rigid pendulum upright, with a rotary encoder reading the angle and nothing telling it where the cart is.

MY ROLE

Designed every 3D model on the rig, Sourced all parts, optimized the electronics layout, and helped tune the PID controller.

OUTCOME

A closed-loop system that holds the pendulum upright and recovers from disturbances.

SolidWorks | Arduino (C++) | PID control | 3D printing

SCHOOL MACHINE DESIGN

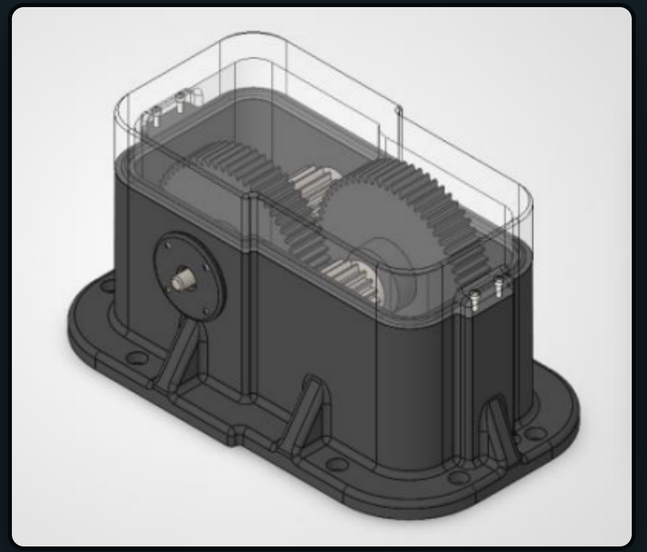
Two-Stage Gear Reducer

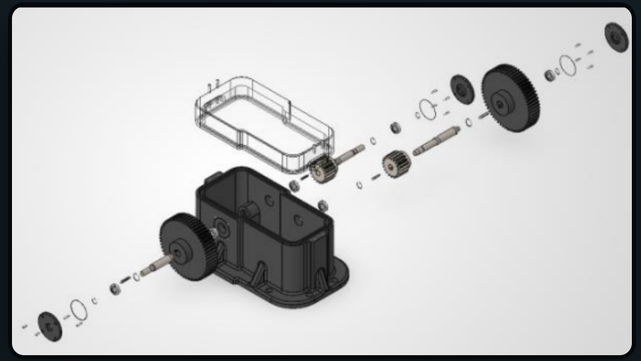
A 9:1 reducer for an industrial conveyor drive, taking 6 hp at 400 rpm down to 45 rpm.

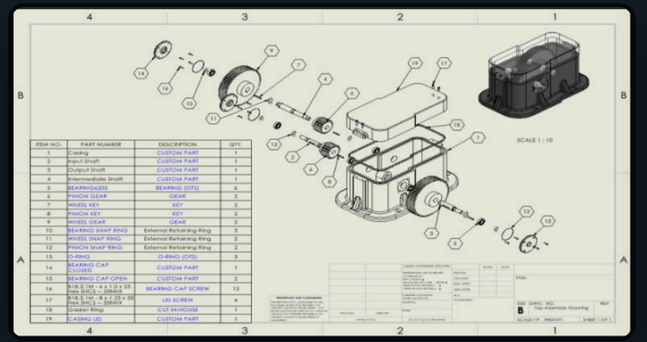
MY ROLE Sized the two-stage gear reduction and ran the bending and moment failure analysis. Designed every part in the assembly except the shaft.

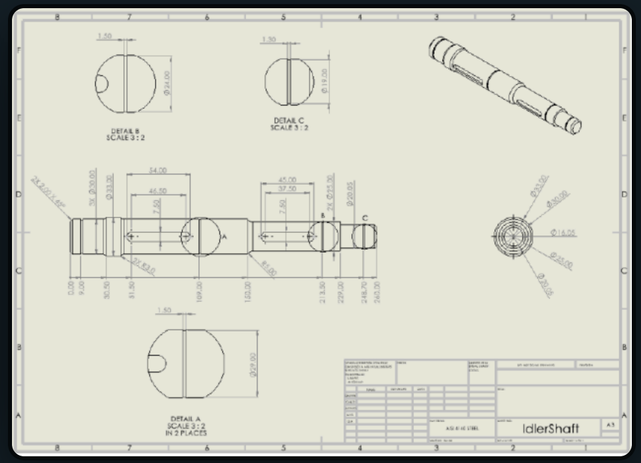
OUTCOME A sealed unit validated for a five-year service life, delivered with a full drawing package.

SolidWorks · Gear design · Failure analysis · Engineering drawings









SCHOOL MCG 2101

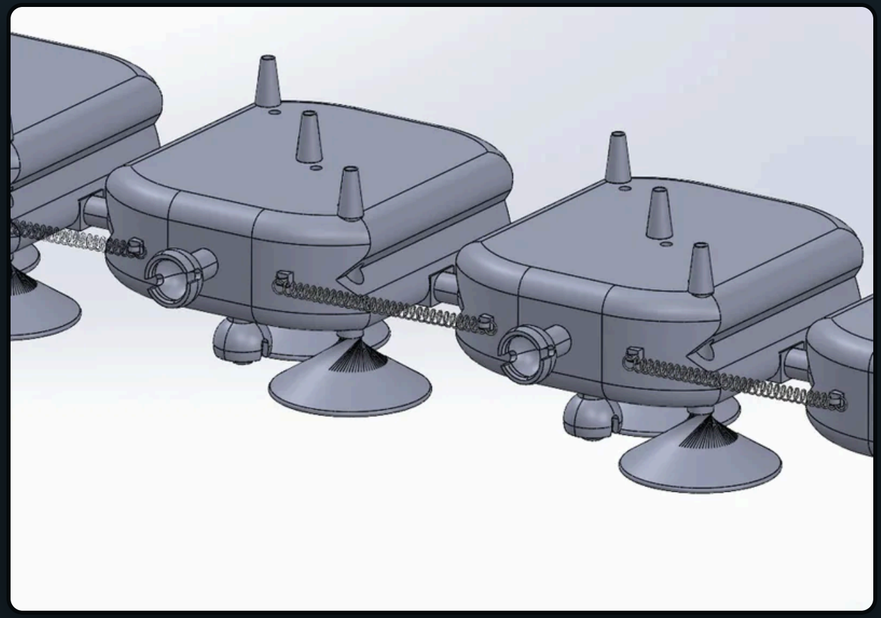
Turbine Blade Mold Polisher

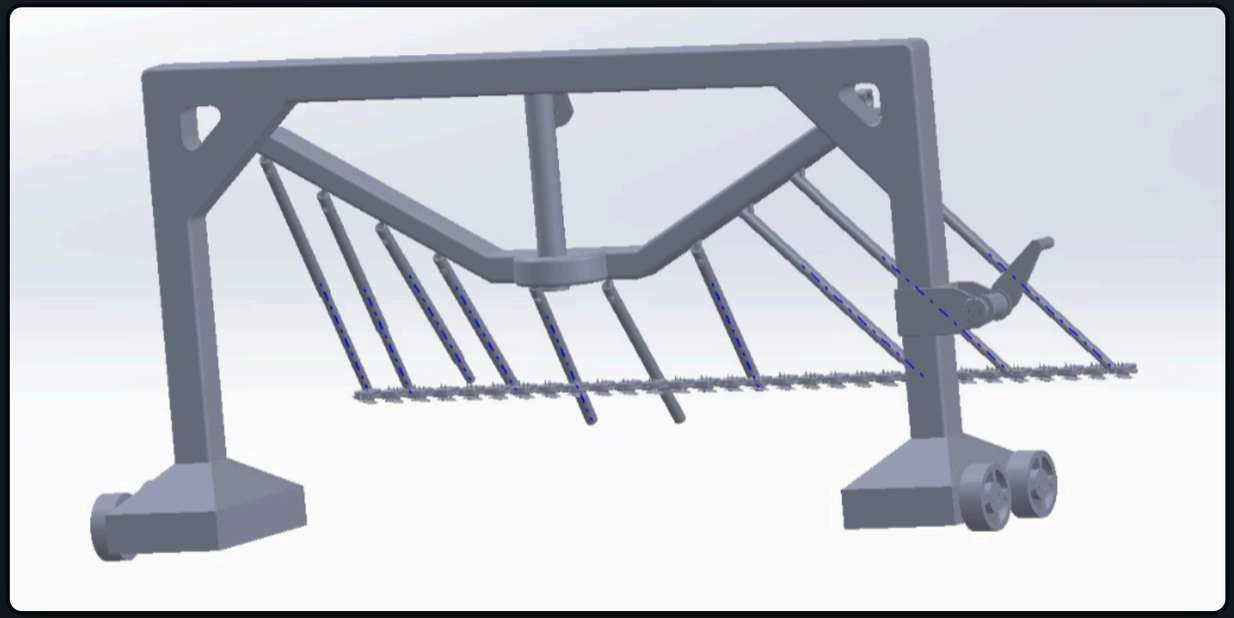
A pneumatic gantry that polishes 9-metre wind turbine blade molds automatically, replacing a job that used to be slow and hard on people's arms.

MY ROLE — Designed the mechanical layout and motion system: the gantry, the yoke, and the spring-connected plates that follow the curve.

OUTCOME — A complete CAD assembly, a full drawing package, and an interactive 3D model that holds constant pressure across the surface.

SolidWorks · Engineering drawings · Pneumatic actuation





THANKS FOR LOOKING

Contact me

I'm open to co-op and internship roles and happy to talk through any of these projects in more detail. Email is the best way to reach me.

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LinkedIn

My Website



HONORABLE MENTIONS

Builds & Repairs

Things I've built or fixed on my own time, mostly because paying someone else to do it felt like a missed opportunity.



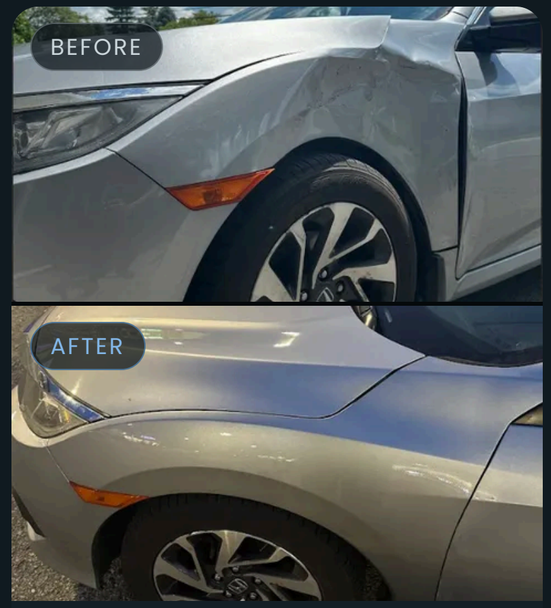
Pool Heater

Found a burst pipe in the heat exchanger, sourced a new one and swapped it in myself.



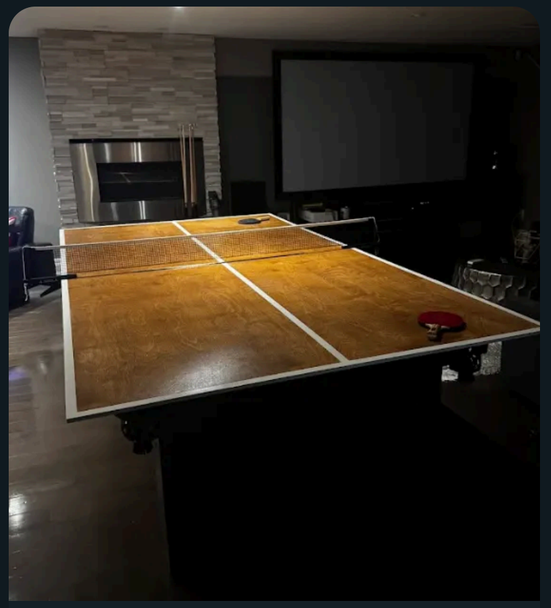
Hot-Tub Heater

Traced a hot tub error to a burnt heater port, then desoldered and replaced it on the board.



Car Fender Repair

Filled and sanded the cracked panel, matched the paint, refit it myself.



Ping-Pong Top

A fold-away top that turns our pool table into a ping pong table in seconds.